Passing primitive data types as parameters will pass a copy of th the method. It is a one way passing in which any changes of the v method (which receive the data) will not affect the source variab calling part). The following program shows the example of Pass-b [*Penghantaran data berjenis primitif sebagai parameter akan m salinan bagi nilai tersebut kepada kaedah. Ia merupakan peng secara satu hala yang mana, sebarang perubahan nilai pada kaed menerima data tersebut) tidak akan memberi kesan kepada pemb sumber (pada bahagian yang memanggil). Atur cara berikut men contoh Penghantaran-Secara-Nilai.*]

```
class SumValue {
  public static void main(String[] args) {
    double num1, num2;
     num1 = 10;
     num2 = 25;
     changeValue(num1,num2)
     System.out.println("num1: "+num1);
     System.out.println("num2: "+num2);

  }
  public void changeValue(double num1, double num2){
     num1 = num1 * 5;
     num2 = 6 + num1;
    }
  }
```

7 Passing object as parameters will pass its address in which any chan the object's value will affect the source object. [*Penghantaran objek sebagai parameter akan menghantar alamat tersebut yang mana sebarang perubahan kepada nilai object akan me kesan kepada objek sumber.*]

2.48 The following program shows the example of Pass-by-reference.
[*Contoh atur cara berikut menunjukkan contoh bagi Penghantaran-secara-Rujukan*]

```
class Circle {
  private double radius,area ;
  public Circle(double r){
    radius=r;
  }
  public void findAreas(Circle r){
    area= radius*2.14*2.14;
  }
 public double getArea() {
   return area;
 }
}

public class PassByRef {
  public static void main(String[] args) {
    Circle circle1 = new Circle(5.0);
    circle1.findAreas(circle1);
    System.out.println ("The area is "+ circle1.getArea());
  }
}
```

The reference of the object is passed

2.49 Returning an object means that an object is returned as the return value.
[*Pemulangan suatu objek bermakna suatu objek dipulangkan sebagai suatu nilai pulangan.*]

```
class Circle {
  private double radius;
  public Circle(double r){
    radius=r;
  }

  public Circle incrRadius(Circle r, Circle s){
    Circle a = new Circle(0.0);
    a.radius = r.radius+s.radius;
    return a;
  }
  }
  public class RetObject {
    public static void main(String[] args){
      Circle circle1 = new Circle(5.0);
      Circle circle2 = new Circle(6.0);
      Circle circle3 = new Circle(0.0);
      circle3=circle2.incrRadius(circle1,circle2);
    }
  }
```

2.50 Java provide a standard practice to name a method as `toString()`. The `toString()` method creates a string that represents the state of an object. It is a no-argument method that returns a string type. It uses concatenation operator (+) to join pieces of string together. The resulting string represents the current state of the object.

```
                 "\n";
        return strMsg;
    }
```

1. Trace the following program code and show the output. Draw a memory layout for the program.
[*Jejak aturcara berikut dengan dan tunjukkan outputnya. Lukis hamparan ingatan untuk aturcara tersebut.*]

```
public class ParameterPassing {
  public static void main (String[] args) {
    ParameterTester tester = new ParameterTester();
    int a1 = 111;
    Num a2 = new Num(222);
    Num a3 = new Num(333);

    System.out.println ("Before calling change:");
    System.out.println ("a1\ta2\ta3");
    System.out.println (a1 + "\t" + a2+ "\t" + a3 + "\n");

    tester.changeValues(a1, a2, a3);

    System.out.println ("After calling change values:");
    System.out.println ("a1\ta2\ta3");
    System.out.println (a1 + "\t" + a2 + "\t" + a3 + "\n");
  }
}

class Num {
  private int value;
  public Num (int update) {
    value = update;
  }

  public void setValue(int update) {
    value = update;
  }

  public String toString() {
    return value + " ";
  }
}

class ParameterTester {
  public void changeValues (int f1, Num f2, Num f3) {
     System.out.println ("Before changing the values:");
     System.out.println ("f1\tf2\tf3");
```

```
        System.out.println (f1+"\t"+f2+"\t"+f3+"\n");

        f1 = 999;
        f2.setValue(888);
        f3 = new Num(777);

        System.out.println ("After changing the values:");
        System.out.println ("f1\tf2\tf3");
        System.out.println (f1+"\t"+f2+"\t"+f3+"\n");
    }
}
```

2. Identify errors in the Program Trial below. You are required to give a brief explanation on each of the identified error and fix the error.
[*Kenalpasti ralat-ralat yang terdapat di dalam Aturcara* Trial. *Kamu dikehendaki menerangkan secara ringkas ralat-ralat yang dikenalpasti dan betulkan ralat-ralat tersebut.*]

```
//Program 2.7
class Trial
{
        private int a;
        private String b;

        Trial(int _a)
        {
                a=_a;
                b=_b;
        }

        public void setAB(int num, String str)
        {
                a=num;
                b=str;
        }
        private int getA()
        {
                return a;
        }

        public void show()
        {
                System.out.println("A: "+getA());
                System.out.println("B: "+b);
        }
}

class TrialTest
{
        public static void main(String[]a)
        {
                Trial ac=new Trial(5,"Five");
                ac.a=10;
                ac.b="Ten";
                ac.setAB(6,"Six");
                getA();
                ac.show();
```

```
            show();
            ac.getA();
    }
```

6: CONSTRUCTOR AND METHOD OVERI

: *KONSTRUKTOR DAN K*